

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

Nov 2013

EE-205 Electrical Machines-I

Date: 12.11.2013, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) What do you mean by electromechanical energy conversion process? Explain in detail [07]
with the help of block representation.

Q - 2 (a) Briefly explain Energy balance and Coupling field reaction. [07]

(b) What do you mean by commutation? What is reactance voltage? Explain the methods [07]
for improving commutation.

OR

Q - 2 (a) Discuss the operating characteristics of DC shunt motor. [07]

(b) 6-pole machine has an armature with 90 slots and 8 conductors per slot and runs at [07]
1000 RPM, the flux per pole is 0.05 Wb. Determine the induced EMF if winding is

I) Lap connected.

II) Wave connected.

Q - 3 (a) Derive the E.M.F. equation for DC machine. What is E_g and E_b ? [07]

(b) What do you mean by torque? Explain armature torque and shaft torque with its [07]
equations.

OR

Q - 3 (a) Discuss the classifications of DC motors along with necessary diagrams. [07]

(b) List out the speed control methods of DC machine. Explain any one method in detail. [07]

SECTION – II

Q - 4 (a) Write and explain: Basic principle and working of Transformer. [07]

Q - 5 (a) Derive the EMF equation for 1-phase transformer. Also explain the transformation ratio. [07]

(b) Briefly explain the cooling system of Transformer. Why it's necessary? [07]

OR

Q - 5 (a) Briefly explain the open circuit and short circuit test. Why it is important? [07]

(b) What is the purpose of tertiary winding in 3 phase transformer? [04]

(c) Why transformer never operates on DC supply? [03]

Q - 6 (a) Describe the NO-Load and ON-Load condition of 1-phase transformer with Phasor diagram. [07]

(b) What is open delta connection? What is its important? [03]

(c) Draw 3-phase winding diagram for following connections (Any two): [04]

I) Δ -Y 11

II) Y-Z 6

III) Y-Y 1

OR

Q - 6 (a) Derive the equations for equivalent Resistance, Reactance and Impedance with use of equivalent circuit of transformer. [07]

(b) Write an equation for voltage regulation of transformer and derive the condition for max. & min. voltage regulation. [07]
